INCH-POUND
MIL-DTL-41820E
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SUPERSEDING
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DETAIL SPECIFICATION

CLOTH, GABARDINE, POLYESTER AND RAYON

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers polyester and rayon spun yarn gabardine cloth.

1.2 Classification. The cloth is available in the following types and classes.

1.2.1 Types.

Type I	- 6.0 ounces
Type II	- 6.4 ounces
Type III	- 8.0 ounces

1.2.2 Classes.

Class 1	- Bleached White
Class 2	- Dyed

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Street, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

AMSC N/A FSC 8305

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2. APPLICABLE DOCUMENTS

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-313 - Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities

COMMERCIAL ITEM DESCRIPTIONS

A-A-59963 - Thread, Polyester

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1, Gray Scale for Color Change
AATCC Evaluation Procedure 2, Gray Scale for Staining
AATCC Evaluation Procedure 8, AATCC 9 Step Chromatic Transference Scale
AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles

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AATCC Test Method 8 - Colorfastness to Crocking: Crockmeter Method
AATCC Test Method 16.3 - Colorfastness to Light: Xenon
AATCC Test Method 15 - Colorfastness to Perspiration
AATCC Test Method 20A - Fiber Analysis: Quantitative
AATCC Test Method 61 - Colorfastness to Laundering: Accelerated
AATCC Test Method 81 - pH of the Water-Extract from Wet Processed Textiles
AATCC Test Method 135 - Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D1683/D1683M - Standard Test Method for Failure in Sewn Seams of Woven Fabrics
ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at https://www.astm.org.)

OTHER PUBLICATIONS

SDL ATLAS

Part Number 402985 – Slub/Knot Replica Set

(Replica set is available for purchase from SDL Atlas Customer Service, 1-803-329-2110.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.crcpress.com.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Unless a specific exemption has been obtained, nothing in this document, supersedes applicable laws and regulations.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Standard sample. Unless otherwise indicated, the finished cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Polyester fiber. The polyester fiber for Type I, Class 1; and Type III, Class 1 shall have an optical brightener. Unless otherwise specified, the use of optical brighteners for Type II (all classes); Type I, Class 2; and Type III, Class 2 is prohibited.

3.4.2 Rayon fiber. A staple rayon fiber shall be used for all types and classes. The use of rayon waste is prohibited.

3.4.3 Yarn. The warp and filling yarn for all types of fabric shall be a blend of staple polyester and rayon fibers. The polyester fiber shall be no less than 60 percent and no more than 70 percent and the remaining fiber shall be rayon. The yarn for all types of fabric shall be spun and twisted into 2 ply yarn.

3.5 Labile sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The dyed cloth shall be “Free” of labile sulfur when tested as specified in Table III.

3.6 Color. The color of the cloth shall be as specified in 6.2.

3.6.1 Visual shade matching. The color and appearance of the cloth for Class 1 and Class 2 shall match the standard sample when tested as specified in 4.7.1.

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3.6.2 Colorfastness. The finished cloth shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.5.

TABLE I. Colorfastness requirements.

Color Evaluation	Laundering Color Change and Staining (min.)	Light (after 40 AFU or 170 kJ/(m^{2}nm)@420 nm) 1/	Perspiration (acid & alkaline) Color Change and Staining) (min.)	Crocking Dry/Wet (min.)
Type I, Class 1 Type III, Class 1	4 2/	4	3-4	N/A
Class 2	3-4	4	3-4	3.5

1/ AFU: AATCC Fading Units
2/ Color change only

3.7 Physical requirements. The finished cloth shall conform to the requirements, listed below, in Table II, when tested as specified in 4.5.

TABLE II. Physical requirements.

Characteristic	Type I	Type II	Type III
Weight, oz./sq.yd. Minimum	6.0	6.4	8.0
Yarns per inch, (minimum) Warp Filling	110 52	110 62	92 40
Breaking strength (pounds), minimum Warp Filling	210 90	210 105	240 110
Weave	2 up-1 down right twill	2 up-1 down right twill	2 up-1 down right twill

3.8 Width. The minimum width (inclusive of the selvage) of the finished cloth shall be as specified (see 6.2).

3.9 Finish. The cloth (all types and classes) shall be processed so as to provide a finish equal to that of the standard sample and meet the requirements of this document.

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3.10 pH. The pH value of the water extract of the finished cloth shall be not less than 5.5 and not greater than 8.5 when tested as specified in 4.5.

3.11 Dimensional stability (all types and classes). The cloth shall not shrink more than 3.0 percent in the warp and 2.0 percent in the filling when tested as specified in 4.6.

3.12 Seam efficiency. The cloth shall have a seam efficiency of not less than 85 percent when tested as specified in 4.6.

3.13 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths of not less than 40 yards. Each length shall be put-up full width on roll as specified in 6.2.

3.14 Fiber identification. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.15 Marking. The face or the back side of the cloth shall be identified by stamping "FACE" or "BACK" on the appropriate side of the cloth at each end of the piece. Markings shall be indelible.

3.16 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.5.3. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.16.1 Toxicity documents. All treatments used to process the cloth shall be identified and accompanied by the appropriate Safety Data Sheet (SDS) information per FED-STD-313.

3.17 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the criteria specified in the contract or purchase order (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

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4.3 First article inspection. A first article, submitted in accordance with 3.1, shall be examined in accordance with 4.5 and tested in accordance with 4.6.

4.4 Conformance inspection. Conformance inspection shall include the examinations of 4.5 and the tests of 4.6, as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits as specified in the contract and/or order, except where otherwise indicated.(see 6.2).

4.5 Yard-by-yard examination.

4.5.1 Visual examination. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with MIL-STD-3064, Type I.

4.5.2 Roll identification and marking examination. During the yard-by-yard examination each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.5.3 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.5.4 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.6 End item testing. The cloth shall be tested for the characteristics listed in Table III. The methods of testing as specified in Table III and 4.7 shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more sample units or the lot average fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. End item tests.

Characteristic	Requirement reference	Test method
Polyester fiber content	3.4.3	AATCC 20A (see 6.4)
Yarn ply	3.4.3	Visual 1/
Labile sulfur	3.5	4.7.1
Visual shade matching	3.6.1	4.7.2
Colorfastness:		
Light (40 AFU or 170 kJ/(m^2nm)@420 nm)	Table I	AATCC 16.3, Option 3 2/
Laundering	Table I	AATCC 61 Test 1A 3/
Perspiration (Acid & Alkaline)	Table I	AATCC 15 3/
Crocking (Wet & Dry)	Table I	AATCC 8 4/
Weight	Table II	ASTM D3776/D3776M
Yarns per inch	Table II	ASTM D3775
Breaking strength	Table II	ASTM D5034
Weave	Table II	Visual 1/
pH	3.10	AATCC 81
Dimensional stability	3.11	AATCC 135, (1), (IV), A(i)
Seam efficiency	3.12	ASTM D1683/D1683M 5/
Toxicity	3.16	4.7.3

1/ One determination shall be made from each sample unit and the result reported as “pass” or “fail”.
2/ Rated using AATCC Evaluation Procedure 1, Gray Scale for Color Change.
3/ Rated using AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.
4/ Rated using AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale.
5/ The needle size shall be 0.040 inch ± 0.001 inch. The needle and looper thread shall be size B polyester yarn in accordance with Type I, Class 1 of A-A-59963.

4.7 Methods and tests of inspection.

4.7.1 Presence of labile sulfur test. In the determination of presence of labile sulfur in textile materials with lead acetate, two (2) 1.50 (± 0.01) gram samples from each material submitted for evaluation shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed in a stannous chloride solution that contains 100 grams of stannous chloride crystals ACS in 100 milliliters of hydrochloric acid ACS (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over top of the test tube. The lead acetate solution contains 5.0 grams of lead acetate CP reagent grade and enough distilled water to make up a 100 milliliter solution; if the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear.

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The test tube containing the textile sample, stannous chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

4.7.2 Visual shade matching. The color and appearance of the cloth shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option C (2019), with sources simulating artificial daylight D75 illuminant with a color temperature of 7500K (± 200) illumination of 100 (±20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

4.7.3 Toxicity test. When required, (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the finish cloth is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If the toxicity requirement (see 3.16) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The cloth is intended for use in the manufacture of clothing items worn by military personnel.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Type and class of cloth required (see 1.2)

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c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1, 4.3)
e. Color required (see 3.6)
f. Width required (see 3.8)
g. Minimum length required if other than specified (see 3.13)
h. Toxicity requirement (see 3.16 and 4.7.3)
i. Conformance inspection and acceptance quality limits (AQL) (see 4.4)
j. Inspection conditions (see 4.2)
k. Packaging requirements (see 5.1)

6.3 Standard sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Replica set. The SDL Atlas Slub/Knot Replica Set is not available from their website at this time. Recommend calling customer service, 1-803-329-2110.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.7 Subject term (key word) listing.
Fabric
Twill
Uniform
White

Custodian:
Army – GL
Navy - NU
Air Force -11

Preparing activity:
DLA-CT

Review activities:
Army - MD
Navy – MC

(Project: 8305-2019- 020)

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